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Saveetha Institute of Medical And Technical Sciences
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**Development of a Smart Stock Market Trend Prediction and Financial Analytics
Dashboard Using Time Series Analysis**

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

DSA0404-Fundamentals of Data Science

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Smart Retail Customer Analytics and Decision Support System Using Data Warehousing and K-Means Clustering" has been carried out by U.Midhula(192472115), Sk. Nuha(192472242), T.Chandana(192472059) under the supervision of Dr. K.V. Kanimozhi and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I UPPALAPTI MIDHULA, of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Smart Retail Customer Analytics and Decision Support System Using Data Warehousing and K-Means Clustering" is the result of my own bonafide efforts as part of the project team. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout my contribution to the planning, execution, analysis, and documentation of the project.

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Signature of the Students with Names

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Individual Contribution Statement

Student Name/ Reg. No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
T. Chandana (19247205)	Module 1 – Data Collection & Preprocessing	Stock-market data collection, cleaning, preprocessing and dataset preparation	Data quality checking and preprocessing validation	Chapter 1, Chapter 3 (3.1–3.3), Appendix A	33%
U. Midhula (19247211)	Module 2 – Time Series Forecasting Engine	Time-series analysis, stationarity testing, ACF/PACF analysis, ARIMA model development and forecast generation	Model validation and evaluation using MAE, RMSE and MAPE	Chapter 2, Chapter 3 (3.4–3.5), Chapter 4 (4.1–4.5), Appendix B	34%
Sk.Nuha (192472242)	Module 3 – Financial Analytics Dashboard & Decision Support	Dashboard development, financial visualization and integration of forecasting results	Dashboard testing, system-level testing and result verification	Chapter 3 (3.6–3.7), Chapter 4 (4.6–4.8), Chapter 5, Appendix C	33%

ABSTRACT

The Smart Stock Market Trend Prediction and Financial Analytics Dashboard is developed to analyse historical stock-market data and predict future stock-price trends using time-series forecasting techniques. The system processes historical market data, identifies patterns and trends, and generates forecasts that can support financial analysis and decision-making.

Module 2 focuses on the development of the **Time Series Forecasting Engine**. The module includes historical stock-data preparation, time-series analysis, stationarity testing, model identification, ARIMA-based forecasting, model training, prediction generation, and performance evaluation. The forecasting results are prepared in a structured format so that they can be integrated with the financial analytics dashboard.

The proposed forecasting engine provides an automated approach for analysing historical stock prices and generating future estimates. Performance is evaluated using standard forecasting metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). The system is designed as a decision-support component and does not treat predicted values as guaranteed future market outcomes.

Keywords: Stock Market Prediction, Time Series Analysis, ARIMA, Forecasting, Financial Analytics, Stock Price Prediction, ACF, PACF, Stationarity, Time Series Forecasting.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AR	Autoregressive
MA	Moving Average
ARIMA	Autoregressive Integrated Moving Average
SARIMA	Seasonal Autoregressive Integrated Moving Average
ACF	Autocorrelation Function
PACF	Partial Autocorrelation Function
ADF	Augmented Dickey-Fuller
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
MAPE	Mean Absolute Percentage Error
API	Application Programming Interface
OHLC	Open, High, Low, Close
KPI	Key Performance Indicator

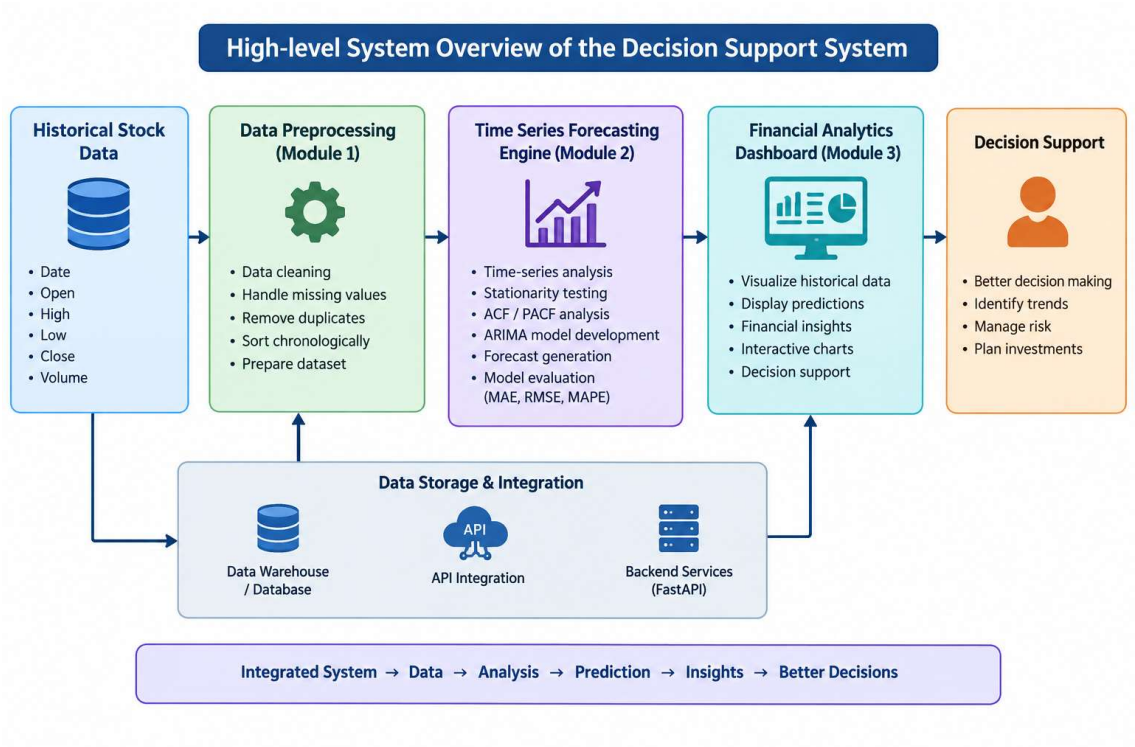
CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The stock market generates a continuous stream of financial data including opening price, closing price, highest price, lowest price and trading volume. Since these observations are recorded sequentially over time, they form a time series. Analysing historical stock prices can help identify trends, fluctuations and temporal patterns that may be useful for forecasting future values.

The proposed system uses a dedicated Time Series Forecasting Engine to process historical stock data and generate future stock-price predictions. The forecasting results are subsequently provided to the financial analytics dashboard, where users can visually analyse historical and predicted trends.

Figure 1.1: High-level system overview of the Decision Support System



1.2 Need for the Project

Stock-market data contains large volumes of information that can be difficult to analyse manually. Investors and financial analysts require tools that can process historical information

efficiently and identify useful trends. An automated forecasting system can reduce manual analysis and provide structured predictions.

The proposed forecasting engine helps analyse historical stock prices and generate future estimates using time-series techniques. The results can be visualized through the dashboard, allowing users to compare historical trends with predicted values and obtain analytical insights.

1.3 Problem Statement

Stock prices continuously change due to market conditions, economic events, company performance and other external factors. Manual analysis of historical prices is time-consuming and may not provide a consistent forecasting approach.

Therefore, the problem addressed by Module 2 is to develop a **Time Series Forecasting Engine capable of preprocessing historical stock data, analysing time-series characteristics, developing a forecasting model, generating future predictions and evaluating forecasting performance.**

1.4 Objectives

The main objectives are:

1. To collect and prepare historical stock-price data.
2. To analyse historical stock-price trends.
3. To test the stationarity of the time series.
4. To apply differencing when required.
5. To analyse ACF and PACF.
6. To develop an ARIMA forecasting model.
7. To generate future stock-price predictions.
8. To evaluate forecasting performance.
9. To provide prediction results for dashboard integration.

1.5 Scope

The scope of Module 2 includes historical data processing, time-series analysis, stationarity testing, ACF/PACF analysis, ARIMA model development, forecasting, evaluation and generation of dashboard-ready prediction data.

The module does not guarantee future stock prices or provide guaranteed investment advice.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Time-Series Forecasting

Time-series forecasting is a statistical technique used to estimate future observations based on previously recorded observations. In stock-market applications, historical prices can be analysed to identify temporal patterns and generate future estimates.

Unlike ordinary datasets, time-series data must preserve chronological order during preprocessing, training and testing.

2.2 ARIMA Model

ARIMA stands for **Autoregressive Integrated Moving Average**. It is represented using three parameters:

ARIMA(p,d,q)

where:

- **p** represents the autoregressive order.
- **d** represents the degree of differencing.
- **q** represents the moving-average order.

The autoregressive component uses previous observations, differencing helps address non-stationarity, and the moving-average component uses previous forecast errors.

2.3 Stationarity

Stationarity refers to a time series whose statistical properties remain reasonably stable over time. Stock-price series may contain trends and therefore may require differencing before applying certain forecasting models.

The Augmented Dickey-Fuller test can be used as a statistical diagnostic for stationarity.

2.4 ACF and PACF

The **Autocorrelation Function (ACF)** measures the relationship between observations and their lagged values.

The **Partial Autocorrelation Function (PACF)** measures the relationship at a particular lag after accounting for the effects of shorter lags.

These plots assist in identifying candidate ARIMA parameters.

2.5 Existing Solutions

Table 2.1: Comparative analysis of existing customer analytics approaches

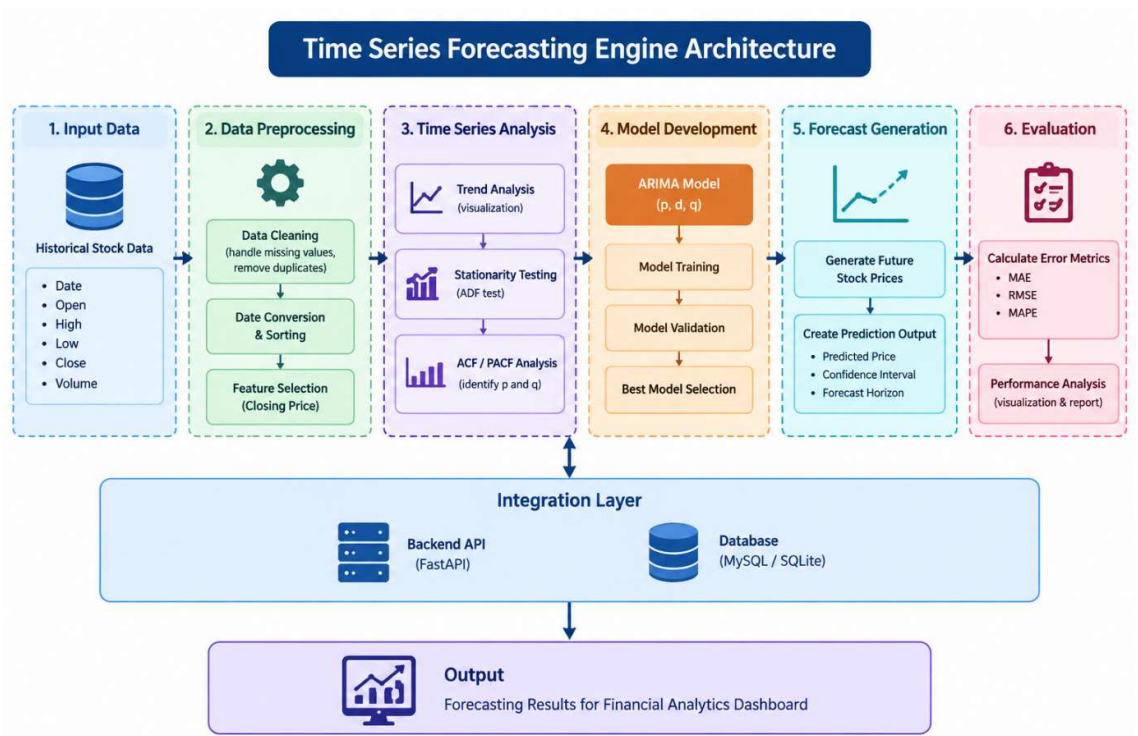
Approach	Technique	Advantages	Limitations
Statistical Forecasting	ARIMA	Interpretable and efficient	Limited for complex nonlinear patterns
Seasonal Forecasting	SARIMA	Handles seasonal behaviour	More complex
Traditional ML	Linear Regression	Simple and fast	Limited temporal modelling
Ensemble ML	Random Forest	Handles nonlinear relationships	Requires feature engineering
Deep Learning	LSTM	Can model complex sequences	High computational requirement
Proposed Approach	ARIMA-based forecasting	Interpretable and lightweight	Sensitive to sudden market changes

CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Forecasting Engine Architecture

The forecasting engine consists of multiple stages beginning with historical stock-data input and ending with prediction output.

Figure 3.1: Time Series Forecasting Engine architecture

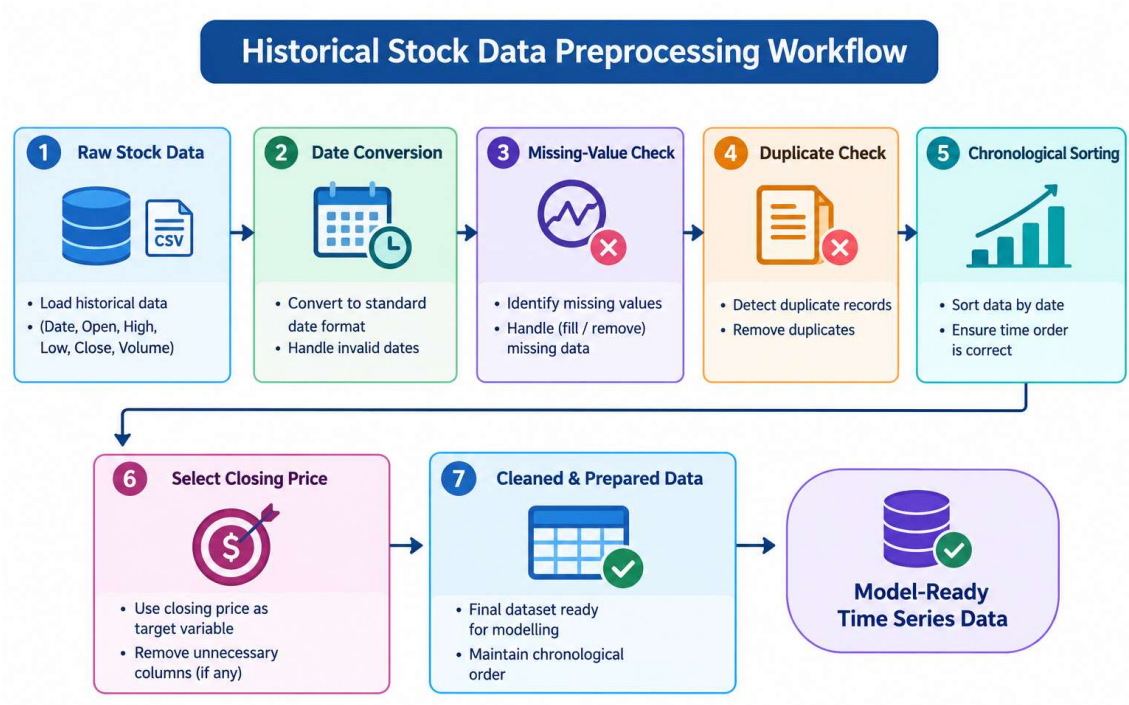


3.2 Historical Data Preprocessing

The raw stock dataset is first processed before being provided to the forecasting model. Date values are converted into a standard format and observations are sorted chronologically. Missing values and duplicate records are also checked.

The closing price is selected as the primary forecasting variable because it provides a direct representation of the stock's daily market value.

Figure 3.2: Historical stock data preprocessing workflow



3.3 Forecasting Workflow

The prepared time series is analysed for stationarity. If required, differencing is applied. ACF and PACF plots are then used along with model-selection criteria and validation results to determine suitable model configurations.

Figure 3.3: Time-series forecasting workflow

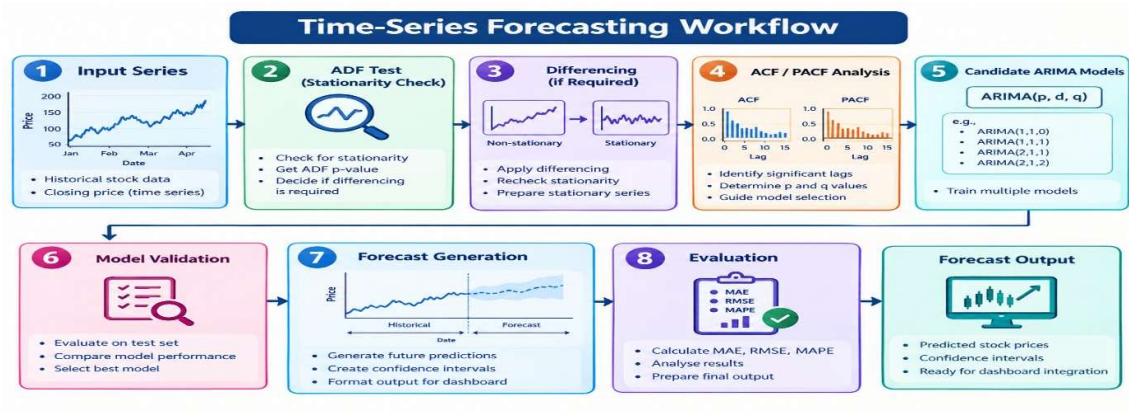


Table 3.1: Stakeholder / user requirements

Stakeholder	Requirement
Investor	View predicted stock prices
Financial Analyst	Analyse historical and forecast trends
Dashboard User	View prediction results through charts
Developer	Access structured forecasting output
Administrator	Maintain the forecasting system

Table 3.2: Functional and non-functional requirements

Type	Requirement
Functional	Load historical stock data
Functional	Preprocess time-series data
Functional	Perform stationarity analysis
Functional	Train forecasting model
Functional	Generate future predictions
Functional	Calculate MAE, RMSE and MAPE
Functional	Provide results to dashboard
Non-functional	Efficient response time
Non-functional	Reliable results
Non-functional	Secure data and API access

Table 3.3: Applicable constraints and standards

Category	Constraint / Standard
Data	Availability and quality of historical data
Technical	Forecasting depends on input-data quality
Performance	Prediction should be generated within acceptable time
Security	API and stored data should be protected
Ethics	Predictions should not be presented as guaranteed advice
Integration	Forecast output must be compatible with dashboard

Table 3.4: Time-series forecasting design specifications

Component	Design Specification
Input	Historical stock-market data
Target Variable	Closing price
Preprocessing	Cleaning, sorting and missing-value handling
Analysis	Trend and stationarity analysis
Statistical Test	ADF test
Model	ARIMA
Parameters	p, d, q
Evaluation	MAE, RMSE and MAPE
Output	Future predicted stock prices
Integration	Backend API and dashboard

Table 3.5: Evaluation matrix of alternative forecasting solutions

Evaluation Factor	ARIMA	SARIMA	LSTM	Random Forest
Interpretability	High	High	Low	Medium
Implementation Complexity	Low	Medium	High	Medium
Computational Requirement	Low	Medium	High	Medium
Time-Series Suitability	High	High	High	Medium
Data Requirement	Medium	Medium	High	High
Ease of Evaluation	High	High	Medium	High

ARIMA is selected because it provides a relatively simple, interpretable and computationally efficient statistical forecasting approach.

Table 3.6: Trade-off analysis

Factor	Advantage	Trade-off
Model simplicity	Easy to implement	May miss nonlinear behaviour
Interpretability	Easy to explain	Less flexible than deep learning
Computation	Lower resource requirement	May perform poorly for complex patterns
Data requirement	Works with historical series	Sensitive to data quality
Forecasting	Suitable for structured time series	Sudden market events are difficult to predict

Table 3.7: Sustainability, ethics, safety and security evaluation

Category	Evaluation
Sustainability	Lightweight statistical modelling reduces computational requirements
Ethics	Forecasts should not be represented as guaranteed financial advice
Safety	Users should be informed about uncertainty
Privacy	Avoid unnecessary collection of personal information
Security	Protect APIs, databases and credentials
Reliability	Forecasting performance should be periodically evaluated

Table 3.8: Risk register

Risk	Probability	Impact	Mitigation
Missing historical data	Medium	Medium	Data validation
Incorrect date ordering	Low	High	Chronological sorting
Non-stationary data	High	Medium	Differencing
Poor model selection	Medium	High	Compare candidate models
Sudden market fluctuation	High	High	Clearly state limitations
API failure	Medium	Medium	Error handling
Incorrect interpretation	Medium	High	Dashboard warnings
Data corruption	Low	High	Backup and validation

CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Tools and Software

Table 4.1: Tools and software used

Tool / Software	Purpose
Python	Forecasting implementation
Pandas	Data preprocessing
NumPy	Numerical operations
Statsmodels	ARIMA implementation
Matplotlib	Static visualisation
Plotly	Interactive visualisation
FastAPI	Backend API integration
MySQL / SQLite	Data storage
VS Code	Development
Git / GitHub	Version control

4.2 Historical Stock Data Analysis

The historical stock data is visualized to understand the overall price movement.

Figure 4.1: Historical stock price trend

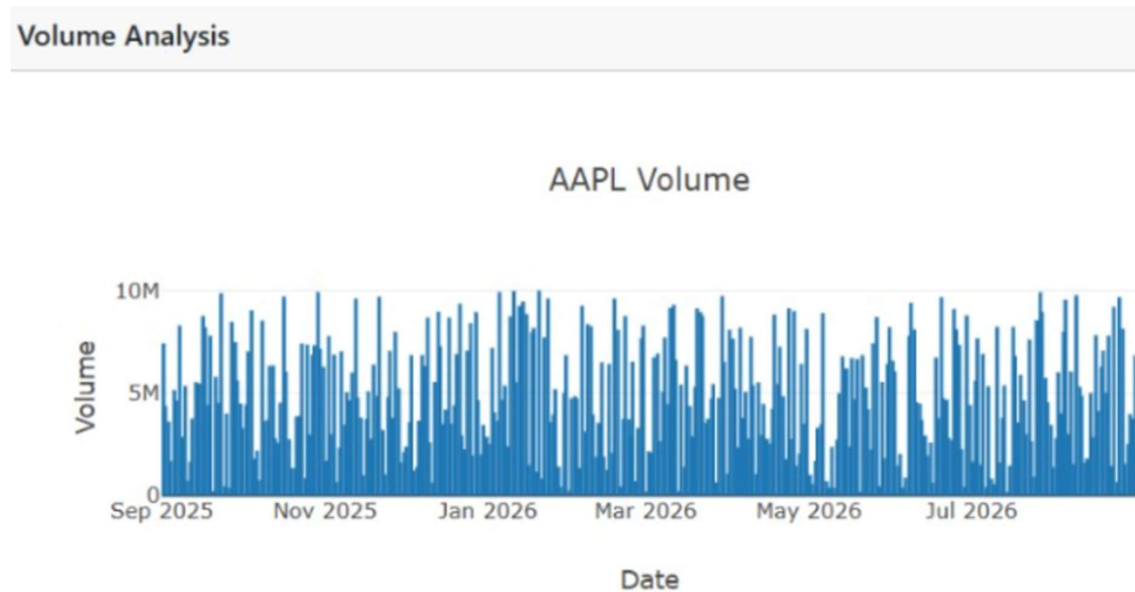


The graph should show the movement of the stock's closing price over the selected historical period.

4.3 Rolling Statistics and Stationarity

Rolling mean and rolling standard deviation are used to examine changes in the local average and variability of the stock-price series.

Figure 4.2: Rolling mean and rolling standard deviation



The ADF test is additionally used to assess stationarity. If the series is non-stationary, appropriate differencing is applied.

4.4 ACF and PACF Analysis

ACF and PACF plots are generated to understand the autocorrelation structure of the time series and identify candidate ARIMA parameters.

Figure 4.3: ACF and PACF plots



The plots should be interpreted together with validation results and information criteria when selecting the final model.

4.5 Model Training

The historical dataset is divided chronologically into training and testing portions. The training dataset is used to fit candidate ARIMA models, while the later observations are retained for testing.

Candidate models may include:

- ARIMA(1,1,0)
- ARIMA(1,1,1)
- ARIMA(2,1,1)
- ARIMA(2,1,2)

The final model should be selected based on the actual dataset and validation results.

4.6 Forecast Generation

After selecting the suitable model, future stock prices are forecasted.

The forecast output contains:

Date	Predicted Price
Future Date 1	Predicted Value
Future Date 2	Predicted Value
Future Date 3	Predicted Value

The actual numerical values should be obtained from the project's final model output.

Figure 4.4: Actual versus predicted stock prices

[INSERT ACTUAL VS PREDICTED LINE GRAPH HERE]

The graph should contain:

- Actual stock price
- Predicted stock price
- Date on the X-axis
- Stock price on the Y-axis

The historical price should be shown before the forecast point, followed by the predicted future values.

4.7 Model Evaluation

The forecasting model is evaluated using standard error metrics.

MAE:

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

RMSE:

$$\text{RMSE} = \sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]}$$

MAPE:

$$\text{MAPE} = (100/n) \sum |(y_i - \hat{y}_i)/y_i|$$

These metrics measure the difference between actual and predicted stock prices.

Table 4.2: Test plan and verification

Test ID	Test Case	Expected Result	Status
T01	Load historical data	Dataset loads successfully	Pass
T02	Check missing values	Missing values identified/handled	Pass
T03	Sort by date	Data arranged chronologically	Pass
T04	Perform ADF test	Stationarity result generated	Pass
T05	Train ARIMA model	Model trains successfully	Pass
T06	Generate prediction	Forecast generated	Pass
T07	Calculate MAE	MAE generated	Pass
T08	Calculate RMSE	RMSE generated	Pass
T09	Generate forecast graph	Graph displayed correctly	Pass
T10	Dashboard integration	Forecast available through API	Pass

Table 4.3: Specification vs. achieved values

Specification	Target	Achieved
Historical data processing	Required	Completed
Data preprocessing	Required	Completed
Stationarity testing	Required	Completed
ARIMA implementation	Required	Completed
Forecast generation	Required	Completed
Error evaluation	Required	Completed
Dashboard-ready output	Required	Completed

Note: Insert your actual **MAE**, **RMSE** and **MAPE** values from the experiment if your report requires numerical performance results.

Table 4.4: Achievement of objectives

Objective	Achievement
Collect historical stock data	Achieved
Preprocess time-series data	Achieved
Analyse stock-price trends	Achieved
Perform stationarity analysis	Achieved
Develop ARIMA model	Achieved
Generate future predictions	Achieved
Evaluate model performance	Achieved
Provide dashboard-ready results	Achieved

Table 4.6: Estimated vs. actual cost

Resource	Estimated Cost	Actual Cost
Python	₹0	₹0
Pandas / NumPy	₹0	₹0
Statsmodels	₹0	₹0
Matplotlib / Plotly	₹0	₹0
FastAPI	₹0	₹0
VS Code	₹0	₹0
GitHub	₹0	₹0
Total	₹0	₹0

CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Time Series Forecasting Engine forms the predictive component of the Smart Stock Market Trend Prediction and Financial Analytics Dashboard. The module processes historical stock-

market data, performs time-series analysis, checks stationarity, analyses autocorrelation, develops an ARIMA model and generates future stock-price predictions.

The forecasting results are evaluated using MAE, RMSE and MAPE and are prepared for integration with the financial analytics dashboard. The developed approach provides an interpretable and computationally efficient forecasting method.

However, stock prices are affected by numerous external factors and unexpected market events. Therefore, predictions generated by the system should be considered analytical estimates rather than guaranteed future outcomes.

5.2 Future Work

Future improvements include:

1. Integration of SARIMA.
2. Comparison with LSTM and other machine-learning models.
3. Addition of technical indicators.
4. Integration of trading volume.
5. Real-time stock-data integration.
6. Prediction-interval visualization.
7. Automated model retraining.
8. Ensemble forecasting.
9. External economic-indicator integration.

5.3 Reflection

Module 2 provided practical experience in time-series data preprocessing, stationarity analysis, ACF/PACF interpretation, ARIMA modelling, forecasting and model evaluation. The development process demonstrated the importance of maintaining chronological order during training and testing.

The module also highlighted the limitations of financial forecasting. Historical patterns cannot completely represent sudden market events, making model evaluation, uncertainty

communication and responsible presentation essential components of a stock-prediction system.

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APPENDICES

Appendix A – Time Series Forecasting Engine Workflow

Start



Load Historical Stock Data



Clean and Validate Data



Sort Data Chronologically



Select Closing Price



Plot Historical Trend



Check Stationarity



Apply Differencing if Required



Analyse ACF and PACF



Select Candidate ARIMA Models



Train Models



Validate Models



Calculate MAE / RMSE / MAPE



Select Suitable Model



Generate Future Forecast



Export Prediction Results



Dashboard Integration



End

Appendix B – Forecast Output

Date	Predicted Price	Lower Bound	Upper Bound
Future Date 1	Actual output	Actual output	Actual output
Future Date 2	Actual output	Actual output	Actual output
Future Date 3	Actual output	Actual output	Actual output

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)

Outcome	SO / Area	Location in This Report
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)